

Re-Making the Makerspace: Body, Power, and Identity in Critical Making Practices

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Abstract

Maker culture is an important part of our contemporary moment, characterized by highly-electronic practices that privilege product over process and the individual contributions of disembodied makers. This article explores the ways making is an embodied and relational process that is present in the practices of people living outside the margins of academic and corporate Maker movements. By showing the ways agricultural production on a farm are embodied processes of making with and through human and nonhuman makers using first-person videoethnographic methods, I extend current critiques of critical making and consider how including everyday, non-electronic material practices in the Maker movement redefines and broadens what it means to make, and therefore what it means to be a maker.

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Introduction

Maker culture is an important part of our contemporary moment. Since Maker Media published the first issue of its DIY-themed magazine *Make*: in 2005, Makerspaces have become recognizable fixtures in many universities as well as community spaces, housing technologies from 3D printers to Arduino boards to audio/visual equipment. In composition and rhetoric, the Maker movement has coincided with an imperative toward **multimodality** (Kress, 2010), where students are encouraged to draw on their outside new media literacies to create compositions “in a new key” (Yancey, 2004). Adopting an orientation toward **“critical making”** (Ratto, 2011), scholars and teachers alike have considered how the fabrication of digital objects engage students in critical understandings of the world. While critical making projects are not exclusively or even wholly electronic, the movement coincides with advances in microcontroller technology, networked objects, and the participatory web. Maker culture calls us back to the DIY (do-it-yourself) movements of the 1980s and 1990s, and it positions us all as producers, inventors, and multiliterate ingenues of the digital age.

But makers have a history that stretches farther back than the first computer, a history that begins with the most traditional of materials: earth, metal, animal fiber, and food (Creighton, 2001; Sohan, 2015; Driskill, 2015; Gabriel & Wagemister, 1997). Such crafts are **socially situated entanglements with materials** (Guéron, 1938/2010; Bratich and Brush, 2011; Parker, 1984) that remix and repeat old tropes (Arola & Arola, 2017; Banks, 2011), taking the form of both written and non-written modes. These made things communicate cultural values and memories (Driskill, 2015) and represent a relationship between process and product (Shipka, 2011). While these “old” technologies are

*intersection between multimodal
composition + critical making*

non-electronic, they are digital in an etymological sense—*digit-al*, meaning of the fingers (Haas, 2007). In these ways, electronic technologies are not definitive of a critical making practice: whether wiring circuits, harvesting crops, knitting blankets, or constructing arguments using pen and paper, we are and always were engaging the body in gathering materials and creating something new.

In this article, I place these non-electronic practices into conversation with the electronics of the Makerspace, following scholars such as Shirin Vossoughi, Paula K. Hooper, and Meg Escudé (2016), as well as Ann Shivers-McNair (2017), in considering how current definitions of critical making are too narrow. This discussion extends from Andrea Riley-Mukavetz, Jacqueline Rhodes, Malea Powell, and Melanie Yergeau's (2016) presentation at the 2016 Cultural Rhetorics Conference, where Malea Powell (2016) proposed an understanding of Maker culture as *Makingspace*, as relational. A Makingspace counteracts the ways that the terms "Maker" and "Makerspace" privilege the *maker* over the *making*, or the relational processes that produce a made thing. While the rhetoric of the Maker movement calls forth ideas about individuals, tools, definitions, and boundaries (Vossoughi et al., 2016), Powell's Makingspace focuses on relations, practices, exchanges, and the embodied ways that humans, nonhumans, and technologies (both electronic and non-electronic) are constantly configuring and reconfiguring one another.

I consider agricultural production a Makingspace, one in which making is present beyond institutional and human-centric boundaries. Agricultural spaces have long been studied as sociological sites of interest and have been the site of much discussion over gendered divisions of labor (Allen & Sachs, 2007; Sachs, 1983). Moreover, the farm and its practices are necessarily embodied and relational—not only to other human laborers and consumers, but to animals, land, weather, and other ambient systems outside the control of human activity. Knowledge on the farm emerges from an embodied relationship with the land (Rios, 2015), with what the land makes and what it allows humans to make. The farm is a network of making practices, both human and nonhuman. Anna Tsing's (2015) novel on the flourishing of the matsutake mushroom economy after the fall of the Pacific Northwest timber industry shows us that while human cultivators are busy harvesting, fixing machinery, making new out of old, and even destroying, nonhumans are making things as well: they burrow up from the ground, transducing sun and moisture and waste into something quite literally fruitful. On the farm, nonhumans contaminate us in an ongoing process of collaborative survival. Making on the farm is a process of relationality—to time, animals, digital and mechanical machines, humans, materials, land and meaning. The critical making that happens in agricultural space is as important to our understanding of makerly practice as the 3D printer, because it shows us the ways everyday people, animals, and things are always and already making in highly complex and relational ways.

In this article, I provide a case study of an agricultural Makingspace by building upon video ethnographic methods developed by Shivers-McNair (2017) and Sarah Pink (2012). Combining Shivers-McNair's first-person, handheld videography with Pink's walking-talking-imaging method for ethnography of the everyday, I consider how participant-makers might teach us what it means to be a maker through a bodily lens. What are the stories that allow us to reconceive of the Makerspace as a Makingspace, as a constellation of practices rather than a set of tools (Powell et al., 2014)? What are the relations that makers engage with as they make? What other knowledges are produced through other relationalities, other material entanglements? In paying closer attention to the embodied stories of makers outside the boundaries of a highly-electronic Makerspace, I work toward a broader and more inclusive definition of what it means to make, and therefore what it means to be a maker.

The Maker Movement

As a community of practice, the Maker movement coalesces around specific shared beliefs and practices that stem from its early days as a large-scale coordination of hackers and tinkerers. In 2005, O'Reilly Media Group premiered its first issue of the magazine *Make:*, the cover depicting a man adjusting his aerial photography rig, with references to DIY, "hacks," and projects that readers could take to their backyard. One year later, *Make:* hosted its first Maker Faire in San Mateo, where more than 10,000 makers and enthusiasts joined together to play, invent, and showcase their wares. As John J. Burke (2014) noted, these beginning moments of the Maker movement were characterized by an organized effort to bring together computing and crafting practices, even though these practices existed prior to the 2000s. Overall, "making" was seen as the process of creating physical objects with digital and Internet-capable tools; it involved woodworking machines, electronics, wearable tech, and even incorporated fiber craft into its communities of practice. It embodied an ethic of ingenuity and inherent human desire to create and was open-source, democratic, and

liberatory (Balsamo, 2009). *Make:* and Maker Faire co-founder Dale Dougherty's (2011) "We are all makers" TED talk exemplified the Maker movement's goals: "Can I do it? Can it be done?"

As Dougherty and other tech giants continued to organize makers through media, fabrication labs, and annual gatherings, scholars were also considering the possibilities that things, and particularly electronically-enabled things, could have for the production of knowledge. "Critical making" (Ratto, 2011) refers to the process of mapping abstract concepts onto physical-technical objects. As a theory-building method, it involves a three-tiered process of construction: first, the maker gathers concepts and materials and decides which ideas can map onto which materials. The second stage involves collaborative prototyping of objects, and the final stage is an iterative process of critical reflection and reconfiguration. Key in this approach is the way makers embed abstract ideas into electronically-enabled objects in an effort to show how society and technology are intimately connected. David Sheridan's (2010) argument for critical making in educational contexts involved the possibility, effectiveness, value, and democracy that critical making brings students. Critical making tools like 3D printers, microcontrollers, and video and audio production are readily available to students to make objects that have meaning and value in our current cultural moment. Not only that, but students have the rare opportunity to take ownership of the processes of fabrication and design. The cultural assumptions of the more popular Maker movement are visible in these visions of new ways of doing scholarship. Critical making involves something inherently *different* than mere fabrication: thought, ingenuity, collaboration, and the freedom to make and explore outside traditional market economies.

While these theories have continued to serve as the grounds for academic Makerspaces and Maker pedagogies, critics of mainstream critical making have pointed out how scholars have failed at meaningful critique of its assumptions. Shirin Vossoughi et al. (2016) demonstrated that along with a same-is-equal mentality across cultural contexts, Makerspaces still operate from contained definitions of making. It involves the use of electronic and Internet-enabled tools, requires explicit rather than tacit reflection, needs to be useful, and is seen as largely being "discovered" and practiced by white men in the technology industry. As such, cultural ingenuity, survivance mechanisms, and art are seen as useless aberrations. Vossoughi, Hooper, and Escudé argued that to make the Maker movement more equitable, educators must set aside definitions of what we *think* is making; technology must be seen as what we practice, what we make with our hands (Haas, 2007).

Vossoughi, Hooper, and Escudé further suggested a historical and cultural perspective on processes of making, where the craft and labor of communities of color, working-class communities, women, and the global South are seen not as mindless rote production but as creative reappropriation of materials (Arola & Arola, 2017; Doris, 2006/2010; Tinari, 2007/2010). Critical making practices that privilege electronic making will always find deficient and resistant makers in communities of marginalized people, requiring "intervention" through new media technology, despite long-standing and deeply multimodal traditions. Riley-Mukavetz et al. (2016) argued that colonization is, in fact, an outcome of an object-oriented maker ideology, one that aims at disconnecting the maker from their makerly relations, elders, and teachers, and erases the ways that marginalized people have always and already been making. While a study of critical making necessarily requires attention to things, it is not enough to *only* concern ourselves with things. Instead, things must be in relation to the people that made them, the history that surrounds them, and the cultures and practices they represent.

Making Bodies and Making Space on the Farm

In following Malea Powell's (2016) critique of the Maker movement and suggestions for more culturally-responsive Makerspaces, I turn to one of my own cultural communities of practice, one of the first places I learned to make: the farm. Agricultural spaces and farm spaces, in particular, require a complex network of making practices to be productive. Beginning from seed, plants are grown through various crop management systems, including fertilization, pollination, and pest management. The products of these labors are then washed and made presentable to a consumer, whether the farmer's own family, market-goers, restaurants, or other buyers. Some plants are processed into other made things: preserves, fiber, cut flowers, etc. Animal husbandry follows a similar process of management, presentation, and distribution, perhaps more steeped in cultures of morality and histories of domestication than the cultivation of plants (Brantz, 2011; Fox & McLean, 2008). Aside from animal and plant products, farmers often are tasked with land management as well: cutting down trees for lumber, mowing, building structures, tilling soil, and working to control invasive species of both plants and animals.

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These processes are material-discursive practices that get carried out through both human and nonhuman interaction and relationality. Bodies are central to these practices: objects emerge as momentarily stable phenomena produced through embodied action and entanglement. In *Making: Anthropology, Archaeology, Art and Architecture*, Tim Ingold (2013) showed how archaeologists formerly perceived certain artifacts as designs imposed on a form. After reorienting their analysis toward the body, the archaeologists found that objects emerged from human, land, and animal bodies: a mound emerges from the ground but is not *built*; a biface handaxe is not *designed* but is the result of a different kind of bodily process. Similarly, making on the farm is not the end result of a farmer's intention placed upon the land; it emerges as a collaborative survival (Tsing, 2015), as the meeting place between human action and "nature's ongoing, material-semiotic intra-actions, actions that may well surprise, annoy, terrify, or baffle humans" (Alaimo, 2007, p. 249). *or perhaps calls the meaning of "design" into question*

In considering agricultural space as emergent from multiple and surprising interactions, we might move one step further to think of making on the farm as emergent from what Donna Haraway (2016) called a multispecies kinship and what Stacey Alaimo (2007) called a contamination. Tsing (2015) showed how sensory collaborations with matsutake mushrooms, or what she called "encounters," are indicative of the ways other organisms exist within us. Smell, in particular, "is the presence of another in ourselves. Hard to describe, yet vivid, smell leads into encounter – and indeterminacy" (Tsing, 2015, p. 45). Whereas Enlightenment and capitalist logics see Nature as something to be imposed upon by the rational human, the farm shows us quite readily that we are contaminated and contaminating, that our boundaries are not set. *expanding our views of who is what constitutes maker/making*
 In this way, the cultural logic of the farm—that animal, insect, dirt, and air will inevitably find their way into our own human materialities—pokes holes into the Maker movement's logic that technological human making is unique and separate from the world-building of other things. When we peer into the homes of bees, we experience a society that is entangled within our own cultural practices: we see the ways our food systems are entirely dependent on their activities of gathering and creating—of their makerly practice. We see the ways they organize their complex societal roles irrespective of us, while still communicating with us, not as other animals, but as *humans* (Armstrong, 2016). But what is most visible to us (or rather, most easily sensed) is what they make: the comb, the propolis, the honey.

Such embodied relationships with the land, flora, and fauna of the farm are what Gabriela Raquel Ríos (2015) described as a way of learning from specific material entanglements. Farmers, like other Maker cultures, share and create knowledge through distributed and embodied networks of literacy- and meaning-making. As an example, county co-operative extension offices, funded and organized through the same land-grant universities that house academic Makerspaces, often hold plant sales, seed swaps, mechanical training, and more. These offices exist as distributed knowledge economies where everyday people can share and receive knowledge about their practice. The extension offices, while funded nationally through the Department of Agriculture, are specifically responsive to communities' demands and agendas (Carroll et al., 2017). A county office in the Appalachian Mountains might host workshops on timber management, whereas an office in the coastal plains may provide guidance on flood management for farms.

These knowledge economies are sites for localized, situated knowledges (Haraway, 1988) and circulate what Ríos (2015) called land-based literacies, "literal *acts* of interpretation and communication that grow out of active participation with land. While these literacies are predominantly extradiscursive, they are nevertheless rooted in relations among discursive phenomena" (p. 64). Leanne Betasamosake Simpson (2014) wrote of the land being "both context and process" (p. 7), the place where learning happens but also the process *by which* learning happens. Simpson argued that Eurocentric ideas about material production, in particular gathering sap, are rooted in isolationist tactics: by extracting the process of syrup production from the land itself, we lose an important part of the knowledges the land holds. Further, Shawn Wilson (2008) demonstrated the role that land plays in staging the production of knowledge. Relationality does not simply entail the people one's research is about; it is also the land, the ancestors, the cosmos or spirit, and the ideas of those we research and learn from. Land is central to literacy because it "carr[ies] the resonance of things" (Brewer & Dourish, 2008, p. 965) and allows us to access past and present simply by engaging with its material. While these scholars are specifically speaking of land, cultivated or not, their arguments extend to farm-based literacies that are also carried out through active performance with and through the environment.

Lastly, farms are spaces where culture and power is reified, resisted, and upended by both human and nonhuman bodies. Bodies on the farm are porous (Tuana, 2007), and while technology is largely constructed as durable and stable (Wajcman, 2004), the machines of the farm co-conspire with us. Farmers coordinate and collaborate with the tractor, the backhoe, and the irrigation canals, marking each other's bodies and actions in the process. But there are histories associated with what it means to be on the farm and what it means to be marked in specific ways. Carolyn Sachs (1983) studied the practices of female farmers across the late 19th and early 20th centuries, finding that farming is purposefully

constructed through technological and institutional systems to be a masculine activity. In 1914, the [Smith-Lever Act](#) established separate cooperative extension offices for men and women, and in 1917, the [Smith-Hughes Act](#) created vocational education initiatives that separated men's agricultural knowledge from women's "homemaking" knowledge. Simultaneously, advances in tractor technologies made the work of the field (men's work) more lucrative while work closer to the home like dairy, eggs, and gardening (women's work) more tedious. Additionally, racist laws of the Reconstruction era relegated farmers of color to share-croppers rather than land holders, subsuming their agricultural practice and production to that of white families. All of these factors help mark Othered bodies as unbelonging ([Khanna, 2012](#)) on the farm. A non-white, non-male body, in relation to an agricultural machine, is made unrecognizable by that machine. However, while these institutional practices create the ruling relations of bodies on the farm ([Smith, 2005](#)), the everyday utterances of farmers are much more fluid. A woman's role is and always has been invisible and yet integral to the economic stability of the farm, with labor being divided in terms of needs rather than roles ([Sachs, 1983](#)). It is in this way that agricultural space serves as a counterpoint to the logics of the Makerspace: how are machines marking bodies, and how are bodies resisting those definitions ([Shivers-McNair, 2017](#))?

Videovoice Methods for Studying Critical Making

In considering agricultural production and place as a Makingspace, this study focuses on the site of my family farm, Summerset Holler, located in Peletier, North Carolina. It is a largely uncultivated plot of 22 acres, with one field and two bee hives as its main source of production. It is managed by my mother, Wendy, whom I have chosen as the main participant in the study. She has been the sole responsible caretaker of the land since my father passed away in 2017 and has spent the past year learning from community members and trial-by-error about land management and apiculture, or beekeeping. While the farm is categorized as a hobby farm by production standards, Wendy's economic stability and food security in the wake of her husband's death is largely dependent on the success of this operation. Wendy is a longtime maker: a knitter, a photographer, and now, a beekeeper and farmer. She is certified through the local beekeeping association and has been tending to her hives since May 2017.

I used a first-person videoethnographic approach to researching makerly practice called [videovoice](#), used more widely in public health action research ([Catalani et al., 2012](#)). The method has its groundings in photovoice methodology and builds from the work of [Shivers-McNair \(2017\)](#) as well as [sensory ethnographers](#) in the field of anthropology ([Pink, 2012](#); [Ingold, 2013](#)). In this method, I have adapted Shivers-McNair's "3D interviewing approach" in which the researcher observes the multimodal practices occurring within a Makerspace by using a handheld GoPro camera attached to her body. The 3D interviewing approach accounts for the movement of bodies along x, y, and z axes and allows the researcher to pay attention to the ways embodied, spatial, and technological boundaries are produced. The approach also acknowledges the ways that the research apparatus helps mark those boundaries, showing how the researcher's own body is implicated in the first-person videographic method. Shivers-McNair triangulates language-based meaning making in the Makerspace with other acts of making, focusing on what possible bodies are made in the Makerspace and *how* they are made through human and nonhuman interaction. 3D interviewing draws from [Pink's \(2012\) "walking-talking-imaging" ethnographies](#) in which the research of everyday life is necessarily situated within place and practice. Pink's sensory ethnographies walk *with* participants along their paths of living and making. They view practice and place as relational and continual and as a route of embodiment that can be *re-experienced*, not analyzed, through video techniques. Both Pink's and Shivers-McNair's multimodal approaches to ethnography are what [Ingold \(2013\)](#) and [Sikand \(2015\)](#) argued is a [shared anthropology](#): a *withness* with data and with participants that requires us to pay attention to movement through worlds, to the ways worlds grow up from the ground, rather than as frozen moments of stability.

This methodology borrows heavily from new media and digital ethnographic methods, which acknowledge both the role that digital media plays in our everyday lives and the ways the digital has the possibility of imaging our experiences in more immersive, networked ways. There has been little scholarly research on the technical literacies and digital media uses of farmers, so the imposition of the digital into an agrarian space may seem abrupt or imposing. However, much has been written about the ways novices experience learning through informal means. [Jennifer Sano-Franchini \(2015\)](#) discussed the ways that East Asian women engage YouTube as a learning interface to decide whether to get double eyelid surgery, and [Lehua Ledbetter \(2018\)](#) addressed YouTube's beauty community as a learning community and a place for procedural discourse. [Matt Morain and Jason Swarts \(2011\)](#) noted, "Students frequently turn to YouTube to fill their own learning gaps" (p. 6). Farmers, too, are using YouTube and video interfaces to create, share, and

obtain new information. YouTubers like Justin Rhodes and Barnyard Bees have huge followings and use first-person videography to share new techniques with participatory audiences. While institutions like the NC Farm Bureau and Southern Sustainable Agriculture Working Group have official channels, it is significant that individual farms and farm workers are sharing their knowledge *outside* of institutional settings. In fact, the inspiration for videovoice methodology is more the work of my father than scholars who use digital videography to understand the world; in the wake of his death, we found several GoPro videos he had produced of his beekeeping practice. His notebooks indicated that these videos were meant to be uploaded to a YouTube channel; he was an avid YouTube viewer and wanted to participate in such a networked sharing of information.

The *witness* at the center of these approaches perform what Powell et al. (2014) called a cultural rhetorics approach to research. *Using story as the central unit of analysis, cultural rhetorics research examines the relationships between humans and their histories, material practices, and ways of telling, to understand how meaning-making happens within the context of cultural communities of practice. Constellation, therefore, is the method of a cultural rhetorics practice that situates stories within their relationships to people, things, land, and ideas (Wilson, 2008). Story is the meeting place for these numerous elements that must be attended and responded to with care. Story is self-authorized, grounded in experience, shows a process of thinking, is exploratory, and is situated in specific time and place (Monberg, 2008). It is concrete, unavoidable, and requires listening and response in ways that traditional ethnography (an act of telling rather than an act of listening) may not.* Further, story as methodology prompts us to enact what Celia Haig-Brown (2003) called a different kind of listening. She considered “what happens when a research participant. . .takes us into the space of her own story. If we want to participate as fully as possible with her telling, then we adjust our assumptions and listen differently” (p. 429). “Opening space” (Haig-Brown, 2003, p. 417) for participants to tell their stories requires a specific kind of orientation toward impossible knowledge—knowledge that we as researchers have yet to build space for in our own knowledge production. Instead, story lays bare the constellations that produce impossible knowledge.

While Shivers-McNair’s and Pink’s work documented everyday practice through an embodied perspective, that perspective is still fully possible knowledge—the knowledge of the researcher, not of the maker. Videovoice ethnography, on the other hand, places the camera in the hands and on the heads of makers themselves. The method draws from established community-based photovoice methodology developed by Caroline Wang and Mary Ann Burris (1994) and Caricia E. Catalani et al. (2012), which asked participants to take photographs and videos of their surroundings and experiences in order for researchers and participants alike to better understand community issues. Typically, participants discuss their image-making choices with researchers in qualitative follow-up interviews. Image-making is treated as a perceptual undertaking where the participant-photographer looks through a lens at their embodied experience of their community. Not only does this allow participants to reflect on their communities and how their communities are framed, but the images serve as a meaningful way to engage many different stakeholders, including the community itself. Similarly, Emma Rose and Alison Cardinal (2018) have theorized a family of user experience methods, called participatory video (PV), that is “motivated by an ethical imperative to empower users to make positive change in their lives, their communities and their experiences with technology” (p. 11). The authors’ use of video diaries during participant commutes provided context to lived experience and “allow participants to take us into their worlds” (p.14), placing agency in users rather than in designers of users’ experience.

Similarly, videovoice participants record video data of their multimodal practice using a small handheld video camera attached to their bodies. As a means for studying critical making practices, videovoice allows participants to reflect on their practice according to their own attention, goals, and interests rather than the researcher’s—a storytelling method that follows along the path of making and illuminates the constellations that a maker exists within. Additionally, first-person handheld videography allows participants to direct the researcher to what is significant. In reflecting the cultural logics of many making communities that center an elder, mentor, or master in the sharing of knowledge (Vossoughi et al., 2016; Dunbar-Hester, 2014), the participant is sovereign over her own making, teaching the researcher what it means to make in a particular relationship with space and place, material, and other humans and nonhumans. Millie Creighton (2001) used this approach to study Japanese craft among middle-class women, comparing her method to apprenticeship, an act of research that is centered in the researcher’s own learning. In combining Pink’s walking-talking-imaging ethnography with an apprenticeship model for research, videovoice positions the maker as master, with the researcher following the steps the participant traces in her makerly practice. In walking the embodied path of the maker, the researcher creates a relational responsibility between herself and her subject; she is a novice in this space, learning *with* and *from* and *within* the participant’s own experiences of making.

In this study, both Wendy and I recorded video of our agricultural practice over three months during the busiest season of apiculture: early summer. During this time, beekeepers build their hive boxes, purchase and install new queens, and work to prevent disease within the new colony. Early summer honey is often drawn off of the upper hive frames to give the colony room to grow—a new colony will produce significantly during this time, growing up to three times its size. During this particular season, Wendy had lost two hives after several late-season freezes, so her goal on the farm was to prepare for receiving two new queens and to begin managing the land for summer growth.

Important to note is that while Wendy places great significance in her ability to do farm work, as she sees it as a means of security: she is still full owner of the property, which is a privilege many farm workers, especially farmers of color, do not have. According to [Megan Horst and Amy Marion's \(2019\)](#) analysis of national survey and census data from 2012–2014, about 36% of landowners are women, whereas only 2% are people of color. Principal operators of farmland were 14% female, but only 6% non-white. White farmers tended to generate the vast majority of income from land and farm ownership, while Black, Asian American, Indigenous, and Hispanic farmers were more likely to rent land or work as laborers. While there is an immediacy to Wendy's work, the privilege of ownership, stacked considerably in her favor in relation to other disadvantaged farm workers, allowed for the slower, more reflective approach we took together in this study. This reflects back onto some of the established critiques of the Maker movement: who gets to have ownership over the means of production? Who gets to farm for its own sake? Who gets to feel agency as a farmer and over farmer technologies? While a critique of power and control in Maker movements is necessary, the same critique is relevant within environmental and agriculture studies and potentially precludes such a slow methodology, as those most severely impacted by power and labor structures do not have time, or perhaps even are not allowed, to be reflective.

In preparing for follow-up interviews with Wendy, I developed questions pertaining to the videos she and I had both produced and asked her to review her own videos prior to our meeting. While choosing what to ask and how to ask it, I considered the ways that Wendy's and my videos framed our agricultural practices in surprising and disruptive ways. I drew from [Kathleen Stewart's \(2007\)](#) theory of *ordinary affects*, in which significant events in the practice of everyday life are defined as surges that jump out from familiar places and circulate in search of new potential. Stewart argued that the everyday forms subjects out of this tension between familiarity/immanence and potential/rupture. Studying the everyday, therefore, is a study of how affects jump out at us, snap us to attention, and form our identities through their circulation (p. 39). In developing my interview questions for Wendy, I looked for visible traces of these surges and how they were situated within broader relations of activity and feeling: emotional moments, bodily moments when a person burped or tripped, frustrating moments, moments when I was surprised by the ways either of us or passersby reacted or acted. I also considered what it meant to pay attention to nonhuman surges: when a bee buzzed against the camera, or a new break in the creek caused us to change the direction of our walking path. I also asked her about her previous makerly history in order to contextualize what she saw in the videos with her broader personal and cultural experience as a maker.

This interview prompted Wendy to develop connections between the relationships and events she was already paying attention to through her video framing strategies. I asked her questions that I would ask myself as an ethnographer, and thus enacted a relational accountability closer to apprenticeship than a researcher drawing out experience from a participant. My questions created space for her to be an analyzer of her own experience. In reflecting on the interview with Wendy, I approached my analysis with the same search for relationality and ordinary affect: What were the moments in the interview where Wendy made connections to her relations, not only her human relations but also to her body and the immediate environment? What was surprising about her experience as a farmer and as a maker? What had she tried, failed, forgotten about? What had clearly laid dormant for her, but was present and living in her experiences on the farm? Most importantly, what had she taught me and others, the apprentices to her work? The categories that follow are the answers to this final question of what I as researcher have been taught, and they show the ways that relationships, not necessarily tools, are integral to an understanding of critical making.

Critical Making at Summerset Holler

Hidden Labor

Labor on a farm is complicated. Throughout history, agricultural processes have been supplanted by products due to increasing availability of technology and increasing investment in globalist-capitalist economies. As consumers,

Connections to
 Star's (1999) call to
 investigate infrastructure
 + what is revealed or
 made visible by investigating
 everyday or mundane
 practices or systems

we only see the “downstream” product of farm labor: the canned goods, the cuts of meat, or the vegetables for sale at market, if we are privileged enough to live in a robust food system. If not, farm processes are even further erased by grocery supply chains and state and federal regulations. But these made things, like any other, carry the traces of agricultural work: dirt still clings to them, marbling in the steak shows the cattle’s eating and movement, bruises on the tomatoes show their handling, and truncated carrots show the depth at which the soil was dug. These traces are indicative of the hidden labor human animals, nonhuman animals, and other living things that invest in their making practices.

In my interviews with Wendy, as well as the videos she and I both produced, that hidden labor becomes much more visible through a videovoice method. Walking-talking-imaging first-person videography allows us to re-experience the processes that result in how food or other agricultural products look, feel, smell, and taste in specific ways. It shows the ways made objects on the farm are the results of our bodies tracing on our materials and illuminates the work (literally the force exerted on a material) involved in making on the farm. In one instance, Wendy notes that there is a disconnect between what she sees on the camera and her phenomenological experience of working with the bees:

I move my head way too much for that GoPro. I just felt like it was, I was moving my head and talking to everybody, or attending to something. I’m constantly moving. I just felt like by watching the video, that I’m frantically moving about, but I don’t feel that way in the moment. What I feel like in the moment is I’m trying to execute a process. (Gollihue, personal communication, October 8, 2018)

While Wendy frames this feeling as a preconceived goal she has when going to the land to work, that goal is not necessarily an end-product. It is a process that is made visible through the video she produces. She notes many times that she has learned something about her beekeeping practice by re-experiencing the steps she took when placing the new queens in the hive. But what she also recognizes is that those steps are much more fluid and granular than doing one action, then another, then another. By understanding how her body moves, what she orients toward and may remember “seeing,” she sees how her body is tracing a path with the land.

An additional laboring that Wendy enacts on the property is that of emotional labor. By taking on the physical labor that her late husband once carried out, she must also face a very personal, intimate form of grief. When Wendy mentions physical labor, she is in the same moment re-tracing the emotional labor she felt at the death of her husband:

All the walks we’ve always done all the way around, and the talks that we’ve had. The plantings, some of those plantings, the mushrooms, they come out and those were the. . . work of his and my hands together on those logs. I’m slowly doing my own thing out there, and I think maybe one day it’ll be a time capsule of just me. But I don’t walk across that area and remember that day as much as finding onions and garlic still coming up a year and a half later from your dad. (Gollihue, personal communication, October 8, 2018)

Walking the outer perimeter of the land, keeping up with mowing, mourning the loss of her two hives, and her concern over the new hives are all wrapped up in the loss of her husband. While the videos do not explicitly walk-talk-image her grief, when she re-experiences them in our interview, there is rarely a moment when she is not tying what she sees to the hidden emotional labor that gets carried out in her work. This grief is not only the loss of a husband and a father to her children, but one of having to re-learn her role as a woman in society.

In viewing the processes of her making and of the making that has been ongoing at the farm, Wendy is able to articulate the emotional labor of loss and responsibility. She sees herself both as she was then and as she is now, and the land is a visible trace of that labor that she sees in the videos.

Bumping Up Against the Nonhuman/Collaborative Survival

What also became evident through my interview with Wendy and watching the videos were the ways humans related and coordinated with nonhumans. This took the form of interactions with the land and with the bees. Notably, Wendy’s discursive understanding of her relationship with the bees was an upending of typical domestication narratives: she related to the bees because *they* “captivated” or “captured” *her*. She clearly articulates the way that nonhuman living things on the farm call her to act in certain ways, to follow them in certain ways. When asked about her relationship with the bees, she speaks of them as her mentors and teachers. The bees have knowledge of land and material that Wendy seeks to learn from and with:

When I stand next to that society of the bees, I feel like I'm, you know, I'm—if they could speak, I would be asking them questions like, “What do I need to do next?” I'm not—I just feel, you know, beneath that society for sure, and that world out at the property and all that it offers up, I really truly, it's bigger than me. (Gollihue, personal communication, October 8, 2018)

In the videos themselves, Wendy and I both “bump” against the nonhuman world constantly. The GoPro's microphone is sensitive to close-range sounds, so when a bee lands on the camera, the buzzing is visceral. When the camera scrapes against the hive box, a vehicle, or another human body, the sound is harsh. What is important about this is that we don't have to be *looking* at the nonhuman world to sense it: our relationality is through nonverbal, nonvisible, non-linguistic means. When Wendy considers what the bees might want to communicate to her, it is a bodily directive: slow down. It is specifically when handling honey that Wendy feels this directive, when working with the bee-made things that she sees are her responsibility. This back-and-forth **affective exchange with nonhuman animals** is a kind of “tracking” or “tracing” (Benson, 2011) that Wendy enacts during this makerly practice. Because her making practice overlaps with the colony's, she sees it as her responsibility to pay attention to the ways the bees coordinate her and their own actions.

Additionally, there are ways in which the land itself collaborates with me and Wendy to create a deeper understanding of the soil-based making that happens on the farm. In each session, we begin recording at the front of the property, where a field has been cleared for several years and which we have been working to build biodiversity into the soil. In one video, we walk along, our cameras each pinned to the ground, watching our steps. What we find are serendipitous plants that had emerged as a result of our concerted efforts as well as some of our mistakes. We stop, stoop down to the ground, handle a bud of red clover that we planted the season before, and stand up again, surveying the whole field. Sometimes, the camera gives a little heave, as our bodies communicate that we are finally seeing the result of our collaboration with the land. Other times, we are simply walking and find something new, something unintentional that popped up from the ground. Garlic seeds from an old field 30 meters away have found their way to the front, and have sprouted scapes. The land is making a new world out of something that Wendy and I never intended, but we are drawn to it as we walk, stopping to examine and learn from the intricate pod that came from the ground's own practice.

These nonhuman constellations that we are drawing are emergent, not from a particular goal that we have set in our minds, but a walking-talking-imaging with the land. Every video shows how Wendy and I begin at the front of the property and find our way, somehow, to the back, stopping countless times to pay attention to something the land has popped up for us to pay attention to. The bees direct us in the same way, showing us when we've overstepped our bounds and flying in from one direction so that we see where they are traveling for food. In this way, the farm and our relationship to it is emergent, relational, and, most importantly, responsive and educational. It teaches us through our nonhuman interactions how to treat it, how to walk along, upon, and beside it.

Time and Rhythm

Significantly, Wendy's awareness of how time functioned on the farm was striking. There were many moments in the videos where one or both of us remarked on the traces that my father had left on the property. Wendy described this as the land being a “time-capsule” that she reconnected with each time she walked the property, a term that is reminiscent of Ríos's argument that land offers a “different orientation to history” where “space, in this case, produces time rather than vice versa” (p. 68). Wendy noted the following:

When I walk across the bridge, that's a time piece for me. He made it with his hands, as badly as it was made, he made that with his hands and his sweat, and by himself. I don't think anybody helped him, I know I didn't. . . I walk across that bridge and go up on that knoll, I'm always, almost always go to a year ago [when he died]. (Gollihue, personal communication, October 8, 2018)

In one video, we are clearing a brush pile together, and I find some old onions that my father had planted that had never grown. When we clear away the grass and find the entire row coming up, we remark on how he would often throw plants in the ground, regardless of appropriate care or season, and they would pop up eventually. It is in this way that the land is a layering of making histories: the bridge we walk over is built on the broken-up foundation of an old concrete factory; there are traces of my father's old gardens and projects everywhere; and where we composted his funeral flowers is the same place where my mother mentions a patch of irises that have popped up. The past carries

~ Relevant
for
David &
Julie
Wohn's
horse
sensor
project

into the present in these moments and creates an experience of time that is all-encompassing and relational, a sensing of multiple times simultaneously.

Similarly, time seems to work differently in different places on the property and through different activities. The videos of Wendy and I conducting a “dress rehearsal” of putting new queens in the hives are very choppy and quick. Their rhythms are confined to a very small part of the land and to very specific activities that are scaffolded. On the other hand, some videos feel much longer, much more drawn out. We move from the front of the property all the way to the back, stopping to pay attention to something that pops out at us. In these videos, we have no real direction but to *be on the land*, and the practice that results is much more visibly ongoing and emergent.

Expertise

Finally, expertise was a common topic of conversation during Wendy’s interview and was present in the human interactions present on the GoPro videos. Wendy was consistently unsure of her expertise on the farm. While she identified as a maker in her relationship to the bees, she identified as a novice in the broader knowledge economy of the farm:

I really feel like I’m still novice. There’s just so much more to learn. I see it out there, I see the stuff that needs to be done, but there’s—you know, there’s certain things that I don’t know how to do yet, and so I really feel like in that whole space out there, I’m still a novice, even inclusive of the bees. I’m still doing research, I’m still learning a lot about the bees. (Gollihue, personal communication, October 8, 2018)

She mentions other beekeepers that have offered their knowledge to her, referring to the tasks they have already accomplished in their own practice like installing queens or checking brood cells in the frames. While the videos show Wendy and I preparing our own course of action for putting in our new queens, they also show two neighbors showing up unannounced to aid us. One neighbor interjects, while one offers to help put things away or go grab tools that we have forgotten. Wendy remarks on these two neighbors’ different approaches to interacting with her: the more involved neighbor just wants to “talk,” while the less involved neighbor wants to “help.” Despite those differences, she believes that they both have more expertise than she has. When I ask Wendy whether she felt like she knew what she was doing when she watched the videos, she compares her knowledge to others, to her husband and her male neighbors most notably. She stated:

I’m doing this without your dad. I was taking the lead and the reins in something that he knew more about because of his first experiences of doing it. I think it made me a little nervous about being confident about what I was doing. . . [Our neighbor] has done this three or four times, so taking his input. At the same time, I felt so versed in our process [that we had practiced the day before], that in a way, it almost was somewhat irritating, his input. But yet it was vital. I know in previous experience in making spaces that you never discredit someone. They’re there to help you and they have experience. Don’t try to do this on your own, listen. (Gollihue, personal communication, October 8, 2018)

While Wendy felt her inexperience in a discursive way—she needed to research more, she needed to read more, watch more YouTube videos, listen to our neighbors, remember what her husband had done, etc.—she *did* tie expertise to her body. She expressed that it was not until our neighbor began interjecting that she felt doubt about her knowledge of the processes she had already been through. We had embodied those processes the day before during our dress rehearsal, physically walking through the steps, picking up the things we would need to pick up, pretending objects were there that were not. In other moments, Wendy discusses the ways her own body is the most reliable source of information and labor, mirroring Anne Frances Wysocki’s discussion of “our bodies—our primary media” (2012, p. 4), that are constantly emerging within their embeddedness. In discussing land management, she notes that she prefers hand methods to using the tractor:

I want to swing the blade. I don’t have anything that will mow anything around those trees like I want to, the base of the trees. I swingblade those. I’ve always done what I feel like I’m gonna get broke down to do anyway. If all of my mechanical equipment breaks, how am I gonna do this without it? Well, I’m gonna swing a blade. It’s the here and the now. My body’s here now, I can do it now. (Gollihue, personal communication, October 8, 2018)

this could be compared to look at in terms of cognition perhaps

I also see this when she discusses her relationship with the bees. She may talk about her need for a discursive understanding of life cycles, diseases, pests, etc., but when she reflects on the moments she feels most expert at caring for the hives, it is a literacy she enacts learns *through her body* in concert with the bees, their structures, and the pathways she must trace to complete the needed maintenance tasks. In reflecting on “learning” the bees, she said the following:

If I work too fast and I feel a little frantic, like I’ve gotta get in and get out, you know, they read that, I think they read that. There’s times when I watch my hands moving those frames and I’m like, slow down Wendy, just relax, and I’m so nervous sometimes just trying to you know, make sure. . . I’m getting better, I’m getting more of a method, but not to the point that I wanna be. And so that’s when there’s times when like, I think I’ve even told you, I’m gonna go do it alone today. And it’s because I want to become more proficient at going in, moving my frames, you know, checking things out, and doing- doing all my little maintenance tasks, and then getting out and remembering what I did, what frames I moved, you know, what I saw on each hive. So I’m trying to get a method going myself. And that’s when I like to be alone sometimes. (Gollihue, personal communication, October 8, 2018)

Learning her method involves an affective experience of the hives, not one derived from her neighbor’s experience or the books she reads. While Wendy’s knowledge of agricultural practice comes from these external sources, it is her own body moving through the motions that tells her when something is working, is right, is certain. An affective relationship where her body listens to the messages the bees send reflects the ways the bees sense her bodily affect, her fear, or her anxiety.

Conclusion: The Embodied Critical Making Practices of the Everyday

While Summerset Holler is one farm among many, and one ^{place} among many, it embodies the kinds of relational practices that should be more present in our understandings of critical making and Makerspaces. Through videovoice methods, I have shown the ways that farm practices involve handwork, prototyping, and distributed networks of knowledge sharing. While these terms help build many of the principles of the Maker movement that are critiqued here and elsewhere, it is important to recognize that what the Maker movement sees is separate and new. It is and always has been a part of the cultural practices of marginalized communities. Wendy learns and makes with her community; she sees her making as inextricable from those relationships. Moreover, farm work is embodied, continual, and collaborative with other humans and nonhumans in ways that are not always made explicit in Maker communities, where the focus is functionally on electronically-defined tools rather than relationships. In addition to making with her human community, she is learning from and with the land- and animal-based teachers that she cohabitates with at Summerset Holler. Perhaps most importantly, I have shown how bodies are integral to understanding expertise within a making environment, how listening to the stories of our bodies in a moment of making is important to understanding how people see themselves as makers, how they identify as makers. How does Wendy know she is a maker? Because of her hands, she said.

What might we in writing studies take from the stories of multimodal, material practice in agricultural space? While critical making provides scholars in our field with the opportunity to explore process, collaboration, and digital literacy as central tenets of composition, we must also recognize the cultural logics we adopt when we uncritically invite the rhetoric of the Maker movement into our classrooms: a privileging of certain ways of knowing, making, and being and marginalization of others. What kinds of composing—what kinds of making—are lost when we focus entirely on digital forms of invention? By framing the material practices of seemingly non-technological makers like farmers as critical, digital, collaborative, and innovative, we create the opportunity for students from marginal places and identities to see themselves, their communities, and their cultures in the production of academic knowledge. Further, by redefining critical making as a relational and embodied practice, as opposed to a given set of tools, we highlight what I argue are more fruitful resonances between the Makerspace and the writing classroom: that writing like making happens in context, through land, material, and cultural practice.

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